

Unconventional monetary policy: the assessment

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Abstract The unconventional monetary policy measures adopted by the major central banks in the period since 2008 are discussed in this paper. We highlight some important differences between quantitative easing and conventional monetary policy and then evaluate the mechanisms through which quantitative easing may propagate to financial markets and the real economy, drawing on perspectives from monetarist and New Keynesian theory. Additional measures, intended to supplement or strengthen the effects of pure quantitative easing, often termed *unconventional* unconventional monetary policy, are also assessed. In our discussion we relate the various articles in this issue to some of the key research questions posed in relation to unconventional monetary policy.

Key words: unconventional monetary policy, quantitative easing, zero lower bound, asset purchases

JEL classification: E44, E52, E58

I. Introduction

The period since the onset of the financial crisis in 2007 has seen central banks around the world confronted with a new and complex set of challenges. The early phases of the crisis were marked by threats to the liquidity and solvency of systemically important financial institutions, most notably Bear Stearns and Lehman Brothers in the US and Northern Rock, the Royal Bank of Scotland, and Halifax Bank of Scotland in the UK. These threats resulted in the functioning of financial markets becoming severely impaired. As market participants revised their views of the creditworthiness of financial institutions, they became willing to provide funds to the banking sector only at a higher price, if at all. As a result, substantial credit spreads—wedges between short-term central bank policy rates and the rates facing households and firms—emerged. This tightening of credit conditions has been a major feature of what economists now refer to as the ‘Great Recession’, a decline in economic activity across the major economies that has not only been severe by historical standards but also protracted and followed by a sluggish and as yet incomplete recovery.

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In response to these events, major central banks around the world adopted a wide range of unconventional policy measures. The starting point for these operations was a broadening of their traditional lender-of-last-resort (LOLR) functions, which sought to address failures in the markets for inter-bank liquidity and avert an immediate escalation of the problems faced by distressed banks. This was followed by a concerted effort to tackle the slowdown in global economic activity. Starting in 2008, central banks around the world began reducing short-term policy interest rates. For example, on 8 October 2008, as the financial crisis intensified, central banks in Canada, China, the euro area, Sweden, Switzerland, the UK, and the US all undertook a coordinated rate cut. In the UK, the Bank of England's Monetary Policy Committee (MPC) cut Bank Rate by a total of 3 percentage points during 2008 Q4 and a further 1½ percentage points in 2009 Q1. But, faced with the prospect of a deep economic downturn, and with short-term interest rates close to the zero lower bound,¹ central banks judged that further monetary stimulus would be required to meet their objectives. As a result, they embarked on a range of measures that can broadly be described as *unconventional monetary policy*.

Foremost among this set of tools is a policy that is commonly referred to as 'quantitative easing' (QE). Although definitions vary, QE typically involves large-scale asset purchases financed by the issuance of central bank money. These sorts of programmes of asset purchases have been put into practice by the Federal Reserve, the Bank of England, and the Bank of Japan, and are sometimes labelled *conventional* unconventional monetary policy. The key distinguishing feature of QE is that it involves the central bank seeking to directly affect asset prices—for example, longer-term gilt yields—other than a short-term interest rate and doing so by actively varying the size of its balance sheet. In addition, central banks have made use of a wide range of policy tools including as forward guidance concerning short-term interest rates, credit easing schemes and long-term repo operations. These measures can be broadly described as *unconventional* unconventional policy.

The articles in this issue explore a number of issues connected to the use of such unconventional monetary policy measures. These range from theoretical and empirical analysis of how unconventional monetary policy affects financial markets and the real economy—the transmission mechanism—to broader questions concerning the implications of unconventional policy measures for our understanding of central banks' objectives and the relationship between central banks and the fiscal authorities. Roger Farmer's article in this issue (Farmer, 2012) focuses on the actions of the Federal Reserve in the United States. He develops a novel analysis of the rationale for QE in which base money creation and asset purchases signal the central bank's intent to achieve the inflation target and thereby condition private expectations in a way that supports economic recovery in the aftermath of macroeconomic downturns.

The empirical analysis of the impact of QE has generated a vast literature during the past three years, to which three articles in this issue—by Chris Martin and Costas

¹ For example, the Bank of England chose to leave bank rate at 0.5 per cent, rather than reduce it right to the zero lower bound. That was because of concerns about possible adverse impacts of extremely low interest rates on the profitability of the banking sector, and the functioning of money markets—see the March 2009 minutes of the MPC for more details. Woodford (2012) includes a discussion of the technical lower bound for central bank interest rates and the factors that may have prevented full convergence on the technical lower bound.

Milas, Mike Joyce, Nick McLaren, and Chris Young, and Francis Breedon, Jagjit Chadha, and Alex Waters—contribute. [Martin and Milas \(2012\)](#) and [Joyce *et al.* \(2012\)](#) both summarize the key insights from this research. The former takes a critical look at a broad selection of the available evidence, while the latter focuses on the financial market evidence in the UK. One of the salient issues recognized by Martin and Milas is that proper assessment of the impact of central bank intervention requires careful construction of the counter-factual path (absent any form of central bank intervention) for response variables such as long-term interest rates. This problem is taken up in more detail by [Breedon *et al.* \(2012\)](#), who introduce a new means of constructing the counter-factual against which unconventional monetary policy should be evaluated and report evidence for the effectiveness of Bank of England policies when evaluated in that way.

Another argument made by [Martin and Milas \(2012\)](#) is that, while there is evidence of a significant effect on the yield curve from Bank of England interventions, there appear to be diminishing returns, in that recent rounds of QE have induced smaller observed responses. Possible interpretations of this trend are offered in the article in this issue by Charles Goodhart and Jonathan Ashworth, who explore some of the limitations of QE as a means of monetary stimulus ([Goodhart and Ashworth, 2012](#)), and by [Joyce *et al.* \(2012\)](#) who recognize that market anticipation of central bank actions makes detection of their impacts more difficult. The authors use information on the changing maturity structure of Bank of England asset purchases as a means to overcome the identification problems that arise in such cases, and do not find evidence that the impact of QE has changed over time. Finally on the empirical side, Peter Sinclair and Colin Ellis discuss the importance of controlling for global monetary trends, particularly those in the United States, in evaluating unconventional monetary policy measures in the United Kingdom and use factor analysis methods in order to address this issue ([Sinclair and Ellis, 2012](#)).

The other articles in this issue probe broader questions arising from the adoption of unconventional monetary policy. [David Cobham \(2012\)](#) presents a narrative on the evolving roles of the major central banks and identifies the period of inflation targeting as a deviation from a norm in which central banks aim for multiple objectives encompassing both financial and macroeconomic stability. The use of unconventional policy measures is interpreted as one stage in the return to that norm following the financial crisis, and the implications for the future status of central banks are analysed. In another article, Bill Allen interprets QE as a form of debt management that shapes the maturity structure of the net liabilities of the state. [Allen \(2012\)](#) surveys the history of debt management in the United Kingdom, and identifies episodes that are comparable to the recent experience in which the effective duration of public liabilities has been reduced through asset purchases of long-dated assets using newly created base money. Finally, [Philippine Cour-Thimann and Bernhard Winkler \(2012\)](#) discuss the unconventional policy measures adopted by the European Central Bank, in particular the rationale for monetary support via long-term refinancing facilities rather than asset purchases funded from money creation.

In the remainder of this article we set the scene for the detailed analysis of unconventional monetary policy measures that follows in the rest of the issue. In section II we explain in more detail what unconventional monetary policy involves, and how it differs from conventional monetary policy. We also explore the difference between conventional unconventional monetary policy (primarily QE) and an array

of schemes that might be considered *unconventional* unconventional monetary policy. The core of the article is section III, in which we provide an overview of the transmission mechanisms linking QE to financial markets and the real economy. We explain in detail both how asset purchases may increase asset prices, and how higher asset prices may then feed through to increased spending in the economy. We highlight the factors determining the effectiveness of these channels and consider the possible distributional consequences of QE. We then offer alternative interpretations of QE from the perspective of monetarist and New Keynesian theory. The former provides an alternative lens through which to understand the mainstream view of the transmission mechanism of QE, and also sheds light on some issues surrounding the behaviour of money and credit data during QE. In contrast, the latter approach generates the strong result that QE is entirely ineffective. We explain that *irrelevance proposition* and draw out its implications. In section IV, we consider other forms of unconventional monetary policy, discussing how they may work and how they have been deployed by different central banks. In section V, we close the article with a brief summary of our main points.

II. The tools of unconventional monetary policy

Central banks turned to unconventional policy measures once the possibility of further conventional monetary stimulus had been exhausted. In this section, we discuss what is meant by unconventional monetary policy, drawing a distinction between conventional unconventional monetary policy (primarily QE), and unconventional unconventional monetary policy. We explain the mechanics of how QE shocks the balance sheets of each of the parties involved in the transaction, and how it differs from conventional monetary policy. We then outline various forms of more creative unconventional policy.

Conventional unconventional monetary policy

QE involves the monetary authority purchasing assets and injecting broad money into the economy. In doing so, QE tries to directly affect long-term interest rates. That is in contrast to conventional monetary policy, which focuses on setting the short-term interest rate (which in turn affects longer-term interest rates through expectations of future short-term interest rates). This distinction is important because at the zero lower bound, the central bank has no room to further reduce short-term interest rates. And in contrast to the conventional focus on setting the price of money, the means by which central banks have sought to achieve this is by actively varying the sizes of their balance sheets and so the quantity of money.² A consequence is that the balance sheets of all the agents involved in the transaction—the non-bank private sector, the central bank, and the banking sector—are affected. We can consider each of these in turn as a way of illustrating the mechanics of how QE works.

² See Goodfriend (2011) for further discussion of central bank balance sheet management and how it compares to other forms of monetary policy

The ultimate sellers of assets are typically the non-bank private sector, for example insurance companies and pension funds.³ By selling assets such as long-dated government bonds (gilt-edged securities, or gilts), the non-bank private sector's holdings of gilts falls (Figure 1). To pay for these gilts, rather than printing currency, the central bank credits the bank accounts of the ultimate sellers of these assets. QE thereby increases their holdings of bank deposits, and so broad money (Figure 1). The central bank finances its purchases by issuing base money in the form of reserves held by commercial banks. It therefore expands its own balance sheet, with the holdings of gilts matched by reserves (Figure 2). The banking sector's balance sheet also expands as the increased holdings of deposits by the non-bank private sector are matched against the newly created central bank reserves (Figure 3). It is this set of perturbations to balance sheets, precipitated by central bank asset purchases, that leads to portfolio rebalancing and so marks the start of the transmission mechanism of QE that is discussed in section III.

To what extent does QE differ from conventional monetary policy? As Bean (2009) notes, there is nothing unusual about central banks purchasing assets *per se*. Seen this way, QE is 'just a return to the classic policy operation of the textbook: an open market operation. The only things that distinguish the present operations . . . are the circumstances under which they are taking place and their scale' (Bean, 2009). Indeed, the policy of purchasing short-dated government securities and expanding the monetary base—what Woodford (2012) terms 'pure' QE—is exactly what occurs when a central bank conducts an open market operation. The key difference is that QE involves a direct injection of a specified quantity of broad money, rather than influencing its price through variations in the price of base money. Another important difference is that central banks have gone beyond purchasing short-dated government securities, the policy pursued by the Bank of Japan between 2001 and 2006. Both the Bank of England and the Federal Reserve have targeted longer-dated gilts, as well as private-sector assets, including corporate bonds and mortgage-backed securities (MBS). That is because the effectiveness of QE may depend on *what* is purchased, as well as *how much*, as discussed in section III.

Figure 1: Non-bank private sector

Assets	Liabilities
– Gilts	
+ Deposits	

Figure 2: Central bank

Assets	Liabilities
+ Gilts	+ Reserves

Figure 3: Private bank

Assets	Liabilities
+ Reserves	+ Deposits

³ If the assets are sold by the banking sector, then the effect on balance sheets is different. We assume here that the central bank succeeds in its aim of purchasing assets solely from the non-bank private sector.

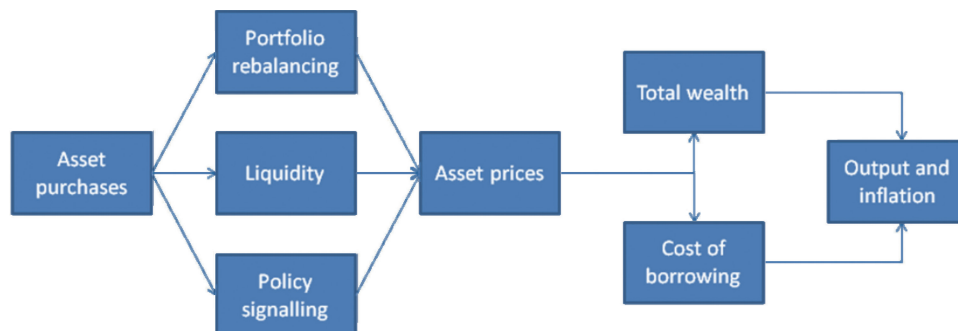
Unconventional unconventional monetary policy

In addition to QE, central banks have deployed a number of other policy initiatives since the onset of the financial crisis. Both the Federal Reserve and the Bank of Canada have engaged in some form of forward guidance, providing information about the future path of policy. A second class of policy initiatives consists of measures designed to improve conditions in the banking sector, ranging from the extended provision of short-term liquidity, to longer-term schemes that provide funding to banks in order to ease credit conditions. In the early stages of the crisis, many central banks took up their LOLR role and extended liquidity support to banks as the availability of funds in private markets dried up. More recently, some policy-makers have provided longer-term liquidity to banks, such as in the European Central Bank (ECB)'s Long-term Refinancing Operation (LTRO), or offered funding as part of schemes designed to boost lending, for example the Bank of England and HM Treasury's Funding for Lending Scheme (FLS). The transmission mechanism for such policies is discussed in section IV.

III. Transmission channels for quantitative easing

In this section we consider the channels through which QE might affect the level of spending and income in the economy. We first discuss how QE might affect asset prices through three channels: *portfolio rebalancing*, *signalling*, and *liquidity* (see Figure 4 for a representation of the transmission channels of QE). In line with central bank communications, we focus on the *portfolio rebalancing* that might ensue following the shock to the balance sheets of the non-bank private sector outlined in section II, and discuss what factors might determine its effectiveness. We also consider how asset purchases may work through *signalling* future policy intentions and through reducing *liquidity* premia. We then go through how a change in asset prices may feed through to nominal spending, highlighting both cost-of-capital and wealth effects. We discuss some of the distributional issues involved with QE, such as the differential effects facing small and large firms, and households with different levels of financial wealth, including pensioners. Finally, we discuss what insights both monetarist and New Keynesian theory might have for our understanding of QE. We explain how a monetarist approach offers

Figure 4: Stylized version of QE transmission channels



Source: Adapted from Joyce *et al.* (2011).

an alternative, but complementary, way of thinking about the transmission mechanism of QE, as well as useful insights into the role of bank reserves and bank lending. In contrast, New Keynesian models in the mould of [Eggertsson and Woodford \(2003\)](#) contain the starkly different result that QE is always and everywhere ineffective. We explain the theory underlying this *irrelevance proposition*, and outline the policy prescriptions that follow from it.

(i) From asset purchases to asset prices

As discussed in section II, the most common implementation of QE has involved the creation of broad money that has been injected into the economy through the purchase of government bonds from investors such as banks, insurance companies and pension funds. This intervention sets off a change in asset prices, which can be thought of as the first leg of the transmission mechanism of QE.

Portfolio rebalancing

The first channel through which QE may affect asset prices is through the *portfolio rebalancing* that it may induce. This channel has been emphasized by the Bank of England's MPC, see [Dale \(2010\)](#) and [Bean \(2011\)](#). By purchasing assets from the private sector, the central bank perturbs the portfolios of the sellers of gilts. In the first instance, the non-bank private sector (for example a pension fund) is left holding money in the form of bank deposits rather than gilts ([Figure 1](#)). We now discuss how this shock to private-sector balance sheets, and therefore portfolios, may trigger a set of adjustments in asset prices.

If the private sector is indifferent between holding gilts and money—they are viewed as perfect substitutes—then the process ends there. In effect, despite the central bank intervention, portfolios remain in balance. At the zero lower bound, money and one-period bonds are both assets that bear no interest and carry little credit risk. As a result, the money created by purchases of *one-period* bonds may be passively absorbed by the private sector. In these circumstances, attempts at this sort of expansionary monetary policy have no impact as the economy is in a liquidity trap. [Woodford \(2012\)](#) refers to such purchases as ‘pure QE’, in so far that the focus is on the quantity of money injected to the economy. The Bank of Japan's policy of QE from 2001 to 2006 is an example of pure QE, in which the purchase of short-dated gilts served principally as a means to inject large sums of money into the economy.

In contrast, QE as practiced by the Federal Reserve and Bank of England has focused on purchasing assets other than short-dated gilts. That is because other assets are likely to be less close substitutes for money. And if two assets are imperfect substitutes, then changes in relative holdings of the two will induce portfolio rebalancing and movements in asset prices, as recognized by [Tobin \(1969\)](#) and [Brunner and Meltzer \(1972\)](#). For example, 10-year gilts are higher-yielding assets than money. So selling these gilts to the central bank and holding on to the money would depress the average returns achieved by investors. And the Fed's later bouts of QE have focused on purchases of MBS, which are likely to be even less similar to money. Further, some investors (such as pension funds) may like to hold long-dated assets in order to match the maturity of their liabilities. Selling gilts therefore also moves them away from their *preferred habitat*

in the maturity structure. Therefore, as the net supply of gilts of a certain maturity is restricted by the central bank intervention, their price increases (and yield falls)—a so-called *local supply* effect.

A second channel through which asset purchases may also affect asset prices is by altering the aggregate amount of interest rate risk in bond markets. The prices of fixed income assets, such as government bonds, are affected by future movements in interest rates, the extent of which is termed *duration*. And if investors dislike this interest rate risk, then they will demand a term premium to compensate them for bearing it. When the central bank purchases long-duration assets, it reduces the aggregate amount of duration risk that remains in the market and needs to be borne by the private sector. As a result, the compensation required by investors to hold all remaining bonds carrying duration risk falls, putting downward pressure on longer-term real interest rates. The Federal Reserve's Maturity Extension programme, often described as 'operation twist', can be motivated by belief in a duration channel (Sack, 2011).

Through these channels, central bank purchases alter the characteristics of investor portfolios. In order to rebalance their portfolios, investors will seek to re-invest the money they hold, searching for alternatives to government bonds which are now more expensive. A natural response is for investors to acquire slightly more risky assets that are now relatively cheaper in comparison to domestic government bonds, for instance high-quality corporate bonds, foreign government bonds, or blue-chip equities. The sellers of those assets will then in turn seek to rebalance their portfolios by holding larger shares of riskier still assets. This process continues until all asset prices have adjusted such that investors, in aggregate, are willing to hold the overall supply of assets.

The strength of any portfolio rebalancing channel depends on a number of factors. First, to what extent do changes in the relative supply of a specific asset held by the private sector cause changes in absolute and relative returns? The idea of a relationship between asset supplies and returns dates back to at least Tobin (1969), and is given empirical support by Greenwood and Vayanos (2010). But it is absent from most modern, microfounded models of financial markets. In these models, the demand curves for financial assets are perfectly elastic and investors do not have preferred habitats. Instead, they care only about the mean and variance of returns. Andrés *et al.* (2004) introduce a group of households (similar to pension funds) who can only invest in long-term bonds, and thereby generate the sort of effects Tobin talked about in a general equilibrium model. However, even then it is still possible that the presence of arbitrageurs—who trade between bonds of different maturities—mitigates the role of preferred habitats. Vayanos and Vila (2009) show that if such arbitrageurs are risk-averse or credit constrained, then portfolio balance effects can prevail. And Yellen (2011) argues that capital constraints may be an important factor limiting arbitrage in the current environment.

Once a role for imperfect substitutability is accepted, the effectiveness of portfolio rebalancing will depend on the degree of substitutability between assets. It is likely to be greater the less substitutable money is for gilts (or whichever assets the central bank purchases—for example MBS) and the more substitutable risky assets are for gilts. In practice, a lack of risk appetite may be a factor limiting investors' willingness to increase their exposure to risky assets. The extremely low yields on government debt that we have seen in some advanced economies may be a result of investors' attaching a safety premium to low-risk, liquid assets. And there is evidence that the flow of funds

has been away from risky asset classes, for instance [UBS Global Asset Management \(2012\)](#) reports a 7 percentage point drop in the average asset allocation to equities among UK pension funds in 2011, to the lowest level recorded since 1974. Other factors may also have constrained portfolio rebalancing. Heightened uncertainty over pay-offs from risky assets, such as equities, will make them less substitutable for gilts. And regulations facing pension funds and insurance companies—such as capital requirements and requirements to hold a certain share of sovereign debt—may limit their abilities to re-orientate their portfolios away from government bonds. One possible response to these constraints is for investors to acquire the debt of foreign governments. That might cause the exchange rate—another asset class—to depreciate. [Yellen \(2011\)](#) notes that falls in domestic interest rates relative to those overseas caused by QE might cause the exchange rate to depreciate, just as is the case with conventional monetary policy.

Signalling

A second channel through which asset purchases may influence longer-term interest rates is through the information revealed about the likely path of future monetary policy. By seeking to loosen monetary policy further, the monetary authority may be *signalling* that it expects policy rates to remain lower for longer, for example by revealing its assessment of the economic outlook. And by undertaking asset purchases, a central bank may demonstrate its commitment to its objectives and its confidence in achieving them at the zero lower bound. This may help maintain credibility and keep inflation expectations anchored.

Liquidity

A third channel—*liquidity*—may operate at times of financial market stress. When financial markets are impaired, investors may demand higher returns on assets to compensate them for the risk there may not be ready buyers for an asset should they wish to sell it. By increasing the volume of trading, and therefore the liquidity of financial markets, central bank asset purchases may be able to bring down liquidity premia. These effects are likely to be present only during the period of asset purchases, and their magnitude may be small in gilt markets, which are normally highly liquid.

Ultimately, the extent to which asset purchases can influence asset prices is an empirical question. Existing research points to clear evidence of declines in government bond yields in the aftermath of central bank asset purchases, as documented in this issue in the articles by Joyce *et al.*, Breedon *et al.*, and Martin and Milas. These articles also report evidence of declines in corporate bond yields, and increases in equity prices. However, there is some disagreement concerning the size and persistence of such effects, and the effects on wider classes of assets are less marked than on corporate bonds. Furthermore, it is difficult to disentangle the different channels—portfolio rebalancing (and, within that, the local supply and duration effects), signalling, and liquidity—through which these effects may arise.

(ii) From asset prices to spending

Higher asset prices should stimulate increases in spending through both reducing the cost of capital and increasing wealth. We explain how a lower cost of capital for

households and firms feeds through to increased borrowing and therefore consumption and investment, and how the functioning of these channels may differ with an impaired banking sector. We note there may be heterogeneous impacts across different sectors, and discuss how QE may still have broader effects on the wider economy. We then consider how we might expect the fiscal authority to respond to a fall in its borrowing costs. Finally, we outline the wealth effects likely to result from a generalized increase in asset prices, and highlight some of the distributional consequences of these effects.

Cost of capital

The rates at which households and firms access finance are typically related to the risk-free rates at the maturity that they seek to borrow. Therefore falls in the yield curve are likely to translate to a reduction in interest rates facing households and firms, in a similar vein to conventional monetary policy. And if banks benefit from increased asset prices in the same way that non-financial corporations do, then their cost of debt issuance will fall. A fall in funding costs, in addition to the fall in risk-free rates, will enable banks to reduce the price of loans to households and firms. This fall in the cost of capital should boost consumption and investment by increasing incentives to borrow and reducing incentives to save.

However, unlike in normal times, central banks turned to QE at a time when the banking sector was badly damaged. Thus, any channels of transmission through the banking sector were likely to be impaired. Accordingly, central banks have focused their communications on the ways in which QE can reduce the cost of capital without needing to go through the banking sector. In particular, portfolio rebalancing towards corporate bonds results in falls in borrowing costs for those companies with access to capital markets. But households and smaller companies do not have access to capital markets, and so may not directly benefit from this channel of QE. This example serves to highlight that changes in the monetary stance—both through conventional and unconventional instruments—inevitably have distributional consequences (for more, see [Bank of England \(2012\)](#)).

Nevertheless, there are still other possible channels of transmission to small businesses. First, there may be supply-chain effects. In sectors in which small firms are connected to large firms via a supply chain, they will benefit either through increased demand (if upstream relative to the large firm) or improved trade credit (if downstream from the large firm). A second possibility is that small businesses in the export sector experience increased competitiveness as a result of domestic currency depreciation.⁴ Third, the falls in risk-free rates and bank funding costs discussed above may be passed on to all of their borrowers. However, if the banking sector remains impaired for other reasons, perhaps because financial institutions are capital constrained, then reductions in their cost of issuing debt may have little impact on the price of credit for households and firms. In section IV of this article we briefly examine some of the recent *unconventional* unconventional monetary policy measures deployed by central banks to stimulate lending in response to problems in the banking sector.

⁴ [Broadbent \(2011\)](#) notes a possible caveat to this real exchange rate channel. While depreciation boosts the competitiveness of small firms, their ability to respond to this stimulus will be limited if constraints on the availability of credit mean that they are unable to expand their operations.

An obvious implication of a fall in gilt yields is that the cost of borrowing for the government is now lower.⁵ How might government spending respond to a fall in gilt yields at longer maturities? We would expect households and firms to increase their borrowing and spending in response to a fall in interest rates. But governments typically behave differently, taking a longer-term view. Their spending plans should therefore be unaffected by cyclical movements in interest rates. QE, like conventional monetary policy, is therefore likely to mainly affect the fiscal authority by reducing the cost of servicing new debt.

Wealth

Increases in asset prices also represent increases in net wealth for their owners. These capital gains should then stimulate spending by households and firms (which are ultimately owned by households). Following the methodology of [Joyce *et al.* \(2011\)](#), we estimate that £375 billion of announced asset purchases to date will eventually provide an overall boost to UK households' net financial wealth of about 30 per cent. These gains will largely have accrued to those holding the most financial assets, in particular older and more affluent households. At the same time, significant proportions of gross household financial assets are held in the forms of pensions. The implications of QE for pensioners and pension providers have attracted much attention. In general, QE affects the value of pension fund assets as well as their liabilities. For example, although annuity rates are likely to fall with gilt yields, this effect should be broadly offset by the increase in the value of the pension pot caused by the very same rise in gilt prices. But some pension schemes, such as defined benefit schemes that were underfunded when QE began, are likely to have been adversely affected by the rise in gilt prices. That is because QE raised the value of the assets and liabilities in similar proportions, implying a widening deficit. Overall, an increase in asset prices is beneficial for their owners, ultimately households. However—as with all monetary policy—there may be winners and losers among different households.

(iii) A monetarist view of QE

We now discuss how a monetarist perspective of QE can provide an alternative, complementary way of understanding the transmission mechanism, and also explain some common misconceptions about how QE works.

For a *broad* monetarist (for example, see [Congdon \(1992\)](#)), QE can be understood as a shock to the money supply, and its transmission mechanism through the prism of broad money supply and demand analysis (for a fuller discussion, see [Bridges and Thomas \(2012\)](#)). As shown in [Figure 1](#), QE positively shocks the money supply by increasing the deposit holdings of the non-bank private sector. For the non-bank private sector to be willing to hold this increased supply of money, one of the

⁵ It is worth noting that by reducing short-term interest rates, conventional monetary policy also reduces the cost at which governments can borrow. However, in so far as QE compresses risk premia on government debt, rather than simply bringing down risk-free rates (which are a function of expectations over short-term rates), there may be an additional impact. The narrowing of the Gilt–OIS (overnight indexed swap) spread seen during QE1 in the UK is consistent with this.

determinants of the demand for money must change. Money demand is a function of its use as a means for making transactions (a ‘medium of exchange’) and as a financial asset (a ‘store of value’). Therefore the demand for money is likely to depend on three variables: the value of transactions in the economy or nominal spending; the overall value of asset portfolios; and the relative rate of return on money as compared to other assets. One of these must therefore change, following a bout of central bank asset purchases. In the first instance, this happens through the familiar process of portfolio rebalancing. The sellers of gilts pass around the money like a ‘hot potato’ as they rebalance their portfolios until the prices and yields of non-monetary assets change so that agents are willing to hold a higher stock of deposits. In this way, a broad monetarist perspective is best seen as an alternative, complementary approach to portfolio rebalancing.

Further, this approach can shed light on some other issues surrounding QE. Importantly, the transmission mechanism relies on *broad* and not *narrow* money. As [Bean et al. \(2010\)](#) stress: ‘In its communications, the Bank of England has stressed the monetary impact of its asset purchases, but that has been on the quantity of monetary deposits in the banking system, not narrow money.’ That contrasts with an approach based on the textbook money multiplier that revolves around the quantity of reserves (narrow, or base money). Under the theory of the money multiplier, a change in the supply of reserves leads to a change in broad money, and nominal spending, many times its size.

This approach has some important implications. First, viewed from a broad monetarist perspective, the null hypothesis would be that the multiplier from QE to broad money is more likely to be unity than a much larger multiple based on historical averages ([Goodhart and Ashworth \(2012\)](#) mention values in the range 10–15). So the failure of broad money to increase by an order of magnitude more than base money is not evidence of a failure of QE. In the event, the money multiplier in the UK has been less than one. [Bridges and Thomas \(2012\)](#) detail the factors—or ‘monetary leakages’—that explain why broad money growth was less than £200 billion during QE1. In particular, they highlight the likely contribution of banks substituting short-term deposits (counted as broad money) for long-term debt and equity (non-monetary liabilities); and of public non-financial corporations (PNFCs) disintermediating from the banking sector by issuing bonds and equity. Both of these factors amount to a reduction in broad money.

Second, under a broad monetarist approach, the behaviour of reserves held by banks is not an important part of the transmission mechanism of QE. It is often claimed that either the high level of reserves held by banks represents idle resources, or that the additional deposits created (as shown in [Figure 2](#)) should be lent out. But, as [Bean et al. \(2010\)](#) state, ‘the level of commercial banks’ reserves *in aggregate* is determined by the way we have funded the asset purchases, not by the commercial banks’ own decisions’. That is, the banking sector in aggregate cannot reduce the amount of reserves it holds, or ‘lend them out’. And the new deposits created are the liabilities held against these assets. So, as [Miles \(2012\)](#) notes, ‘it is not surprising that banks’ reserve balances increased by a lot—it is simply a matter of arithmetic’. Therefore, the only way that more reserves can lead to an increase in bank lending is if it changes the incentives to lend, perhaps by reducing funding costs as discussed previously.

(iv) The irrelevance proposition

So far we have explained how the imperfect substitutability of assets is a key assumption for QE to work through portfolio rebalancing, and noted the absence of these effects in most state-of-the-art macroeconomic models. However, even with imperfect substitutability, a well-known class of models still has no role for any portfolio rebalancing channel. In this section we explain the intuition behind this *irrelevance proposition*, first derived by [Eggertsson and Woodford \(2003\)](#), and its implications for the conduct of monetary policy at the zero lower bound.

The basic contention of the irrelevance proposition is that when the central bank replaces private-sector holdings of gilts with money, it does not, in fact, change the risk characteristics of the private sectors' portfolios as a whole. That is despite money being acknowledged as less risky than long-dated gilts. The reason is that the credit risk associated with the gilts has not disappeared by being moved out of the private sector's balance sheet. It has merely been shifted to the government's consolidated balance sheet. And households ultimately bear any risk taken on by the government through the taxes that they will be susceptible to in the future. As such, purchases of riskier assets make the government's net worth more uncertain, increasing the risk to households of an additional tax burden in the future. Anticipating this, households recognize that the riskiness of their portfolios as a whole remains unchanged, and therefore in balance. This proposition has elements of both the [Modigliani–Miller \(1958\)](#) theorem—in so far as a mere re-shuffling of risk between agents does not change the total amount of risk to be borne—and Ricardian equivalence—as households look through the government's intertemporal budget constraint to anticipate future tax burdens.

[Woodford \(2012\)](#) offers a concrete example to illustrate how this sort of mechanism might work in practice. If the central bank buys MBS, as the Federal Reserve has done, it takes real-estate risk on to its own balance sheet. While households will no longer directly be exposed in the event of a house price crash, the central bank will. Therefore, in the crash state, the central bank's earnings will be lower. And this will result in lower earnings distributed to the Treasury, which will in turn result in higher taxes levied on the private sector. Thus, in aggregate, households' post-tax income is equally affected in the event of a house price crash, whether or not they or the central bank hold MBS.

The assumptions underlying this result (just like those underlying both the Modigliani–Miller theorem and Ricardian equivalence) are very strong. Nevertheless, proponents of this view argue in favour of a different policy response at the zero lower bound to QE. They posit that, as noted by [Krugman \(1998\)](#), with nominal interest rates at the zero bound, the only way to reduce real rates further is to generate an increase in inflation expectations. In order to create expectations of higher inflation, the central bank must make agents believe that monetary policy will be looser than would normally be the case. And looser monetary policy will lead to inflation being above target in the future. So the optimal response to the zero lower bound involves a promise to overshoot the inflation target in the future in order to avoid an even greater undershoot today.

The challenge in implementing this sort of policy is for the central bank to convince the private sector to believe its promise to keep interest rates lower and inflation higher in the future. This is difficult because once the economy begins to recover owing to the reduction in real rates that the central bank's promise has delivered, then there is no incentive for the central bank actually to follow through on its promise. The policy is

time-inconsistent. The central bank will have already reaped the rewards of its promise and will now find it optimal to return inflation to target (see [Meltzer \(2003\)](#) for a discussion of premature exit from loose monetary policy following the Great Depression). And so a promise of an extended period of low interest rates lacks credibility. As such, this sort of monetary policy solution is only likely to be effective if the central bank can find some way to convince market participants that it will overshoot its inflation target by keeping interest rates low for an extended period. One way of interpreting unconventional monetary policy measures such as QE is that they help to achieve this goal, for instance through the signalling channel for future short-term interest rates discussed earlier in this section. Indeed, [Woodford \(2012\)](#) attributes most of the effect of asset purchases to some form of signalling. Alternatively, the central bank may use its communication strategy in order to signal future policy intentions more directly, an approach discussed in detail in section IV.

IV. Unconventional unconventional policies

In this final section we discuss a range of other policy initiatives pursued by central banks in addition to QE. First we consider policies of *forward guidance*, which provide information about the future path of policy and are theoretically appealing in New Keynesian models that feature the irrelevance proposition. Second, we touch upon a broad class of *liquidity* policies that central banks around the world have embarked on. Although these are primarily extensions of central banks' LOLR functions, rather than monetary policy, they have also been deployed to provide funding to banks on a longer-term basis. Third, and finally, we consider some specific schemes aimed at reducing funding costs in the banking sector, which might loosely be considered *credit easing*.⁶ We do not consider any policies that have not been used by major central banks, such as monetary financing or 'helicopter money'.

(i) Forward guidance

In order to implement the New Keynesian prescription of committing to a path of loose monetary policy, some central banks have adapted their communication strategy to incorporate differing degrees of *forward guidance*. As soon as the effective lower bound was reached in December 2008, the Federal Open Market Committee (FOMC) announced that the federal funds rate target was likely to remain unchanged 'for some time'. Subsequently, this language was strengthened to refer to 'an extended period' in March 2009. Then in August 2011 the FOMC specified that its best collective expectation was for the federal funds rate to remain exceptionally low until at least mid-2013, which was extended to late 2014 in January 2012. In April 2009, the

⁶ In this article, we define credit easing as any policy that aims to ease credit conditions by providing funding to lenders. We do not use it—as the term is sometimes defined—to refer to any purchases of private securities, such as corporate bonds or MBS. If these policies are accompanied by an increase in the money supply, then we consider them to be variants of quantitative easing.

Bank of Canada announced a conditional commitment to maintain its policy rate target at 0.25 per cent until Q2 2010. In some ways, this sort of measure is a special case of the communication strategy of regularly publishing interest rate forecasts, as practiced by the Reserve Bank of New Zealand since 1997, the Norges Bank since 2005, and the Riksbank since 2007. But for both the Federal Reserve and the Bank of Canada it represents a deviation from their conventional strategy of refraining to offer any forward guidance.

This sort of forward guidance has been effective in influencing interest rate expectations. Woodford (2012) marshals a range of evidence to show that financial market measures of interest rate expectations generally respond to central bank communications. However, he highlights that financial markets have not fully priced in the cuts implied by the central bank announcements. For example, following the Bank of Canada's April 2009 announcements, OIS rates for 10- and 12-month maturities did not fall all the way to 0.25 per cent, despite the conditional commitment to keep rates at that level for at least a year. And the Federal Reserve's use of forward guidance has been associated with a smaller—though still substantial—effect on interest rate expectations, which Woodford attributes to the Fed being careful only to offer forecasts of the most likely evolution of interest rates, rather than any form of commitment. Overall, Woodford concludes that recent experiences suggest that 'central bank statements about future policy can, at least under certain circumstances, affect financial markets—and more specifically, that they can affect markets in ways that reflect a shift in beliefs about the future path of interest rates.'

But he argues that the policies of the Federal Reserve and Bank of Canada discussed so far are not examples of central banks have using forward guidance to fully carry out the New Keynesian prescription by committing to *overshoot* inflation in the future. Rather, they can be thought of as simply guidance about how policy would be set in the future by a forward-looking inflation-targeting central bank (or a central bank following a Taylor rule). In other words, they did not incorporate a promise to be 'irresponsible' *in the future* based on a zero lower bound constraint that was only *currently* binding. The Federal Reserve's recent policy announcement on 13 September 2012 goes some way towards doing this. The Fed's statement stresses that 'a highly accommodative stance of monetary policy will remain appropriate for a considerable time after the economic recovery strengthens', and extended its projection for the federal funds rate to remain exceptionally low until mid-2015. Yet it still remains some way from a commitment to temporarily higher inflation or explicit guidance about the conditions required to justify policy normalization.

(ii) Liquidity operations

Throughout the crisis, central banks have extended various schemes involving the supply of funds to the financial sector. It is important to note that these have predominantly been presented as *liquidity* operations, fulfilling the role of LOLR rather than strictly monetary policy. The aim of the LOLR function (Bagehot, 1873) is to prevent temporary shortages in liquidity causing the failure of fundamentally solvent financial institutions. The commonly accepted means to achieve this is for central banks to lend on a large scale to solvent institutions at penal rates and against good collateral.

One method of providing such liquidity is through the use of central bank standing facilities, through which financial institutions can access relatively short-term funding against certain types of collateral. During the crisis, major central banks have expanded both the range of assets eligible as collateral and the range of counterparties that can access standing facilities. A second option is to offer large-scale programmes to specific institutions in distress. Examples of this sort of emergency lending include the Bank of England's support to Northern Rock, and the Federal Reserve's extension of credit to Bear Stearns, AIG, and Citigroup. A third set of policies involves one-off (or occasional) central bank operations to provide longer-term financing to the banking sector, typically providing high-quality and liquid securities (such as Treasury Bills) in exchange for lower-quality collateral. An example is the Bank of England's Special Liquidity Scheme (SLS) introduced in April 2008, under which roughly £185 billion of Treasury bills were lent against a range of securities for effectively up to 3 years.

A prominent example of such a policy is the ECB's LTROs. The maturity of these operations was initially set at 12 months but was extended to 3 years in December 2011. In two LTROs held in December 2011 and February 2012, the ECB engaged in gross lending of over €1 trillion for 3-year terms for banks from across Europe. Although the Bank of England's SLS and the ECB's LTRO share operational similarities, they differ in their stated aims. The Bank of England introduced the SLS to improve the liquidity position of the banking system. The ECB has argued, as discussed in this issue by [Cour-Thimann and Winkler \(2012\)](#), that the intention of the LTROs was to fix the monetary transmission mechanism by reducing the spreads between risk-free rates and the cost of funding to banks. As such, they constitute part of the monetary framework.

(iii) Credit easing

Reducing bank funding costs in order to stimulate bank lending has also been the stated aim of policies often classed as *credit easing*. An important feature of the crisis has been a dramatic increase in perceptions of the riskiness of the banking sector. That has manifested itself in high spreads on bank debt, which have in turn put upward pressure on the rates offered by banks on retail deposits. The higher cost of funding for banks has then resulted in higher borrowing costs for households and firms. As discussed in section II, the transmission mechanism of QE largely bypasses the banking sector. Although asset purchases may indirectly reduce bank funding costs somewhat through portfolio rebalancing towards risky assets, central banks have turned to other policies to more directly reduce credit spreads. The ECB's LTRO certainly fits in this camp, and could arguably be classed as either credit easing, a liquidity operation, or a measure designed to assist implementation of the monetary policy framework.

In the United Kingdom the central bank and the government have responded to elevated credit spreads through the introduction of a range of measures intended to stimulate the flow of funds to the banking sector. The first of these was the National Loan Guarantee Scheme announced in 2011.⁷ This scheme aims to reduce the risk to investors who commit capital to the banking sector through government guarantees for up to £20 billion of bank debt. In effect, investors acquiring bank debt up to this limit

⁷ See <http://www.hm-treasury.gov.uk/nlgs.htm> for further information on this scheme.

face the same level of risk as they would do purchasing gilts. And banks receive funding at a discounted rate if they agree to reduce the cost of borrowing for small and medium-sized enterprises (SMEs). However, £20 billion of funding only represents around 8 per cent of gross lending to SMEs by the largest UK banks over the past year, and so the scheme is limited in scale.

A larger scheme aimed at boosting bank lending was unveiled by the Bank of England and HM Treasury in July 2012 in the form of the Funding for Lending Scheme (FLS). Under this scheme the Bank of England will provide funding to commercial banks for up to 4 years—the horizon over which banks typically seek to finance themselves in the markets. At a minimum, banks will be eligible for funding equivalent to 5 per cent of their existing portfolio of loans to households and PNFCs. Five per cent of the stock of existing loans is equivalent to roughly £80 billion across the eligible institutions. A novel feature of the scheme is that additional unlimited funding will be available to banks provided they can demonstrate that any incremental funding is matched by fresh lending to UK households and firms that expands the total size of their loan book. In other words, every pound of additional lending made during a reference period from July 2012 to December 2013 increases the amount that a bank is eligible to borrow under the scheme. This aspect creates strong incentives to boost lending. There are also strong price incentives that encourage banks not to deleverage. Banks maintaining or expanding their loan book will pay a fee of 0.25 per cent—substantially below the market price for comparable funding—on their entire allocation of funds. But, for every 1 per cent fall in lending over the reference period, the fee on the entire drawdown increases linearly by 0.25 per cent, up to a maximum fee of 1.5 per cent for banks that contract their stock of lending by 5 per cent or more. There is also, therefore, a price incentive for a bank to deleverage less than it would otherwise have done. Overall, the scale of the FLS and the incentives embedded within it should relieve any constraints placed on bank lending by elevated bank funding costs.

V. Conclusion

In this article we have reviewed the main elements in central bank efforts to stimulate economic and financial markets since short-term policy interest rates hit their lower bound. The policy of quantitative easing is now considered *conventional* unconventional monetary policy when it involves expansion of the central bank balance sheet through the acquisition of assets such as government bonds. The rationale behind such a policy is that it induces portfolio rebalancing by investors in the direction of private-sector assets that ultimately forces down the private cost of finance and boosts wealth, which should then boost nominal spending in the economy. Our discussion highlighted the key stages in the process of portfolio rebalancing and factors that may limit its extent, including the risk preferences of investors and the regulatory constraints that they face. We also outlined other channels through which the transmission of unconventional monetary policy may operate, namely signalling of monetary policy intentions and improving the liquidity of asset markets. We explained the irrelevance proposition that has been levelled at unconventional monetary policy, whereby households look through the veil of the government's consolidated balance sheet, and therefore do not

need to rebalance their portfolios following central bank asset purchases. A key theme in the literature and in the articles included in this issue, is that when there are limits to private-sector portfolio rebalancing, conditioning expectations over future monetary policy and future inflation outcomes is vital in providing incentives for increases in private-sector spending. In section IV of the paper we expanded on what has become known as *unconventional* unconventional monetary policy. Specifically, we discussed examples of forward guidance for short-term interest rates as a further instrument for guiding private expectations, the liquidity support offered to the banking sector, and the introduction of credit easing measures such as the National Loan Guarantee Scheme and Funding for Lending Scheme in the United Kingdom as a way to stimulate bank lending when bank funding costs remain elevated.

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